

Sol.

(1)

1. $H(f)$ è la convoluzione di tre rect con banda $\frac{1}{2T}$ e quindi ha banda $\frac{3}{2T}$. Il segnale

$\sin\left(3\pi\frac{t}{T}\right) = \sin\left(2\pi\frac{3}{2T}t\right)$ è una sinusoidale di frequenza $\frac{3}{2T}$. Quindi l'uscita alle componenti $V_1 \sin\left(3\pi\frac{t}{T}\right)$ è nulla dato che $H\left(\frac{3}{2T}\right) = 0$.

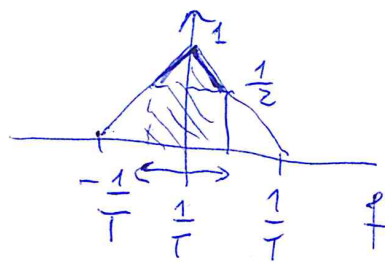
L'uscita di un filtro alle costanti V_0 è V_0 o, e $h(t) = V_0 \cdot \delta(t)$.

Allora $h(t) =$

$$\int_{-\infty}^{+\infty} h_0 \sin^3 \frac{t}{T} dt = h_0 \int_{-\infty}^{+\infty} \sin^2 \frac{t}{T} \cdot \sin \frac{t}{T} dt$$
$$\stackrel{\uparrow}{=} h_0 \cdot \int_{-\infty}^{+\infty} \frac{1}{T} \text{triangolo } fT \cdot T \text{ rect}(fT) df$$

Parseval

$$= h_0 T^2 \left(\frac{1}{T} \cdot \frac{1}{2} + \frac{1}{T} \cdot \frac{1}{4} \right) = \frac{3h_0 T}{4}$$



L'uscita è dunque costante e vale $V_0 \frac{3h_0 T}{4}$.

La potenza di un segnale costante è la costante al quadrato

$$\lim_{T \rightarrow +\infty} \frac{1}{2T} \int_{-T}^T \left(\frac{3V_0 h_0 T}{4} \right)^2 dt = \left(\frac{3V_0 h_0 T}{4} \right)^2$$

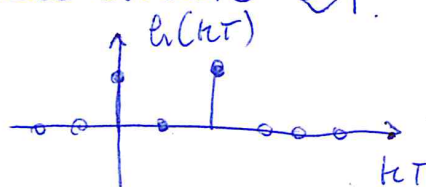
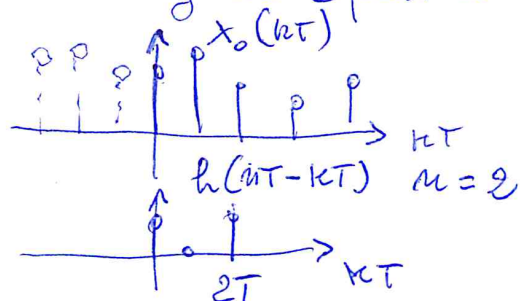
(2)

$$2. \quad H(f) = 1 + e^{-j2\pi f 2T}$$

$$\cos\left(k\frac{\pi}{2} + \frac{\pi}{10}\right) = \cos\left(2\pi \frac{1}{4T} kT + \frac{\pi}{10}\right), \quad f_0 = \frac{1}{4T}$$

L'uscita è $x(kT)$ nulla, dato che $H(f_0) =$
 $= 1 + e^{-j4\pi \frac{1}{4T} T} = 0, \quad H(0) = 2,$
 $y(kT) = 2V_0$

Il filtro è FIR, e le uscite a $x(kT)$ e $x_0(kT)$ sono uguali o parte dell'istante $2T$.



3. Dato che $H_n(f) \cdot H_r(f) = A_n A_r \text{rect}\left(\frac{|f| - f_0}{2/T}\right)$,
 come nello schema ideale, nulla cambia per il segnale
 e si ha

$$\tilde{x}(t) = \frac{1}{2} A_n A_r A_T \cdot x(t)$$

$$M_{\tilde{x}} = \frac{1}{4} A_n^2 A_r^2 A_T^2 M_x$$

$$M_x = \text{area } R_x(f) = \frac{R_0}{T}$$

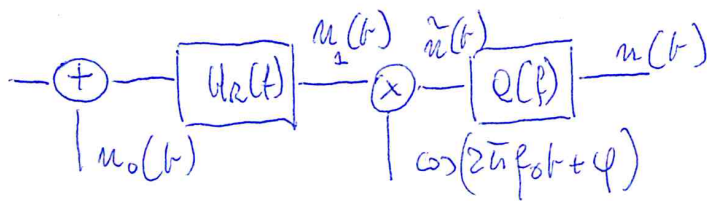
Per il rumore, la densità spettrale dopo il filtro
 $H_r(f)$ risulta

$$R_{m_1}(f) = R_0 \cdot |H_r(f)|^2 = R_{m_1}^{(+)}(f) + R_{m_1}^{(-)}(f)$$

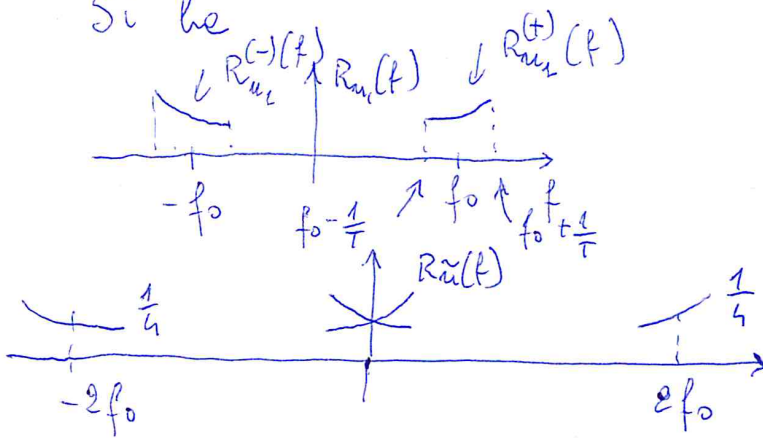
con

$$A=A_r \quad R_{m_1}^{(+)}(f) = R_0 \frac{A^2}{4} \left(1 + \frac{f}{f_0}\right)^2 \text{rect}\left(\frac{f - f_0}{\frac{2}{T}}\right)$$

$$R_{m_1}^{(-)}(f) = R_{m_1}^{(+)}(-f) \quad (R_{m_2}(f) \text{ è pari})$$

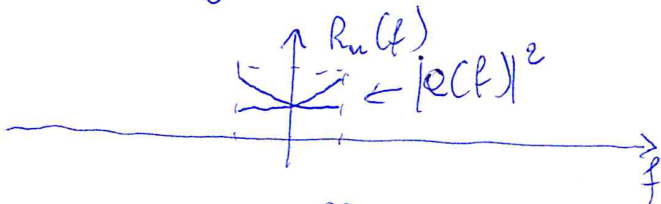


Si ha



$$R_m^{\sim}(f) = \frac{1}{4} R_{m_1}(f - f_0) + \frac{1}{4} R_{m_1}(f + f_0)$$

$$M_m^{\sim} = \frac{1}{2} M_{m_1}$$



$$M_m = \frac{1}{2} M_m^{\sim} = \frac{1}{4} M_{m_1}$$

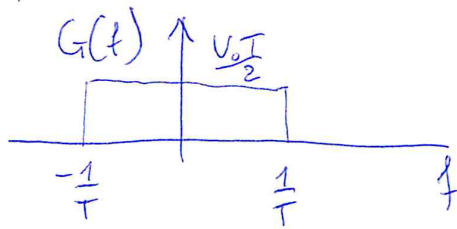
$$M_{m_1} = 2 \cdot \int_{-\infty}^{+\infty} R_{m_1}^{(+)}(f) df$$

$$M_m = \frac{1}{4} M_{m_1} = \frac{1}{2} \frac{R_0 A^2}{4} \int_{f_0 - \frac{1}{T}}^{f_0 + \frac{1}{T}} \left(1 + \frac{f}{f_0}\right)^2 df = \frac{R_0 A^2}{8} \frac{2}{3} \frac{12 f_0^2 T^2 + 1}{f_0^2 T^3}$$

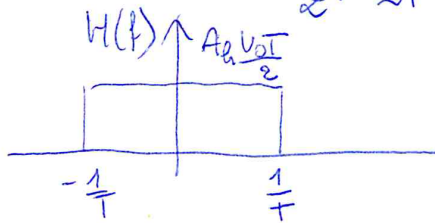
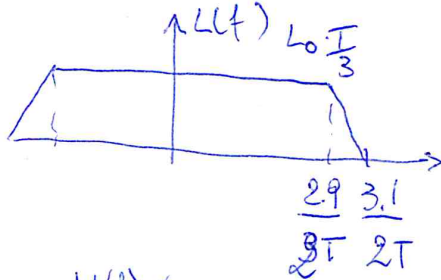
A=Ar

Infine $\Lambda = \frac{M_m^{\sim}}{M_m}$

h.



$$G(f) = \frac{V_0 T}{2} \text{rect}\left(\frac{f \cdot T}{2}\right)$$



$$C(f) = \frac{L_0 T}{3} A_n \frac{V_0 T}{2} \cdot G(f)$$

$$c(k) = \hat{V}_0' \cdot g(k)$$

$$\hat{V}_0' = \frac{L_0 T}{3} A_n \frac{V_0 T}{2}$$

$$c(k) = \hat{V}_0' g(k) \quad \text{e d Nyquist} \quad c(0) = V_0' = \hat{V}_0' \cdot V_0$$

$$\sigma_m^2 = R_0 \int_{-\infty}^{+\infty} |H(f)|^2 df = R_0 \frac{A_n^2 V_0^2 T^2}{4} \cdot \frac{2}{T}$$

$$P_e = \frac{3}{2} Q\left(\frac{V_0'}{\sigma_m}\right)$$

$$P[\hat{e}(kT) = -1, e(kT) = 3] = P[V_0' \cdot e(kT) + n_k \in (-2V_0', 0), e(kT) = 3]$$

$$= P[-5V_0' \leq n_k \leq -3V_0'; e(kT) = 3] = \frac{1}{h} \left[Q\left(\frac{3V_0'}{\sigma}\right) - Q\left(\frac{5V_0'}{\sigma}\right) \right]$$

(5)

5. $e(hT)$ è una v.e. gaussiana a media nulla e
varianza $\sigma_a(0) = 1$

$$P[e^2(hT) + e(hT) - 2 < 0] =$$

$$= P[(e(hT) - 1)(e(hT) + 2) < 0]$$

$$= P[-2 \leq e(hT) \leq 1] = \Phi(1) - \Phi(-2)$$